

Faiq Ahmad Khan

Ph.D. Student, Electrical Engineering | Forward-Deployed AI / LLM Systems | Inference • Post-Training • VLM

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SUMMARY

Ph.D. researcher and hands-on AI systems engineer building multimodal and LLM/VLM pipelines from fine-tuning through local production inference. Experience with QLoRA/LoRA, Megatron-based full-parameter training, DeepSpeed ZeRO-3, quantization, RAG, agent/tool integration, and GPU inference serving. Built a local Qwen agent stack with SGLang, speculative decoding, MTP/DFlash2, LiteLLM, TLS, and agentic tools, increasing measured generation throughput from ~93 to ~250 tokens/s (~2.7×).

TECHNICAL SKILLS

- LLM/VLM & Post-Training: PyTorch, Hugging Face Transformers, PEFT, LoRA, QLoRA, SFT; hands-on VLM instruction tuning; Megatron-based full-parameter training; DeepSpeed ZeRO-3; BF16/INT4 workflows.
- Inference & Serving: SGLang, vLLM, llama.cpp, LiteLLM; speculative decoding, MTP/DFlash2 drafting, KV-cache/inference optimization concepts, tensor-parallel deployment, OpenAI-compatible APIs.
- Efficiency: INT8 post-training quantization, 4-bit base-weight quantization for QLoRA, pruning experiments, gradient checkpointing/accumulation; model-size, memory, latency and throughput benchmarking.
- RAG & Agents: CLIP/text embeddings, Chroma/vector search, reranking, LangChain, structured JSON outputs, tool/function calling, local-model integration with agentic workflows.
- Production & Systems: Python, C/C++, Linux, Docker, Git, REST APIs, TLS/reverse-proxy deployment, distributed training concepts (DDP/FSDP/ZeRO/TP/PP), GPU performance profiling.

SELECTED AI SYSTEMS & RESEARCH PROJECTS

Local LLM Inference & Agentic Deployment — Qwen / SGLang / LiteLLM

- Deployed a local Qwen model behind SGLang and an OpenAI-compatible LiteLLM interface, exposed through TLS and connected to agentic coding/tool workflows.
- Integrated DFlash2 speculative drafting and MTP; benchmarked generation throughput improvement from ~93 to ~250 tokens/s (~2.7×) on the same local inference stack.
- Worked across inference-engine configuration, draft/target model integration, serving, API compatibility, and deployment debugging rather than only offline model evaluation.

Qwen3.5-9B Multimodal Dietary Fine-Tuning — QLoRA

- Built multimodal instruction data pairing egocentric meal images with structured food, portion, calorie, and item-count supervision; enforced JSON-formatted outputs.
- Fine-tuned Qwen3.5-9B with QLoRA (LoRA rank $r=16$; quantized base weights) for dietary image understanding and calorie estimation.
- On a 5-image pilot evaluation, reduced mean calorie absolute error from 43.1% to 22.4% (48% relative reduction) and item-count MAE from 1.20 to 0.40 versus the untuned 9B baseline.
- Compared the tuned model against multiple frontier-model baselines on the same pilot set and tracked validation loss for checkpoint selection and overfitting analysis.

Megatron Full-Parameter Fine-Tuning Study — Small-Data VLM Adaptation

- Ran a Megatron-based full-parameter training experiment on the same dietary multimodal task and analyzed training/validation dynamics against the parameter-efficient QLoRA approach.
- Diagnosed poor generalization in the small-data regime: despite training convergence, held-out calorie error worsened versus the untuned baseline, motivating PEFT/data-scaling choices rather than overclaiming the failed run.

DietAI24 — Retrieval-Augmented Food Identification & Calorie Estimation

- Built sequence RAG for food identification and FNDDS-based calorie estimation using CLIP embeddings, vector retrieval, reranking, and SSIM gating to skip redundant frames.
- Exported per-frame structured predictions for meal-level analytics and evaluated retrieval/generation behavior across egocentric food sequences.

Neo-EEGPT — EEG Seizure Detection & Android Deployment

- Trained an EEG Transformer for neonatal seizure detection with subject-wise validation; deployed to Android with INT8 PTQ, reducing model size by 3.53×.
- Evaluated unstructured pruning and identified the practical limitation that sparsity alone does not guarantee latency gains without sparse-kernel/runtime support.

EXPERIENCE

Graduate Research Assistant — The University of Alabama, Tuscaloosa, AL | May 2024 – Present

- Research and engineer multimodal AI systems spanning VLM fine-tuning, retrieval-augmented vision, model evaluation, efficient inference, and edge deployment.
- Design controlled experiments for model adaptation and deployment, including PEFT vs full-parameter training, structured supervision, quantization, and inference benchmarking.

Lab Engineer — NUML, Islamabad, Pakistan | Mar 2020 – Mar 2024

- Led programming/AI labs; mentored student teams in machine learning, signal processing, embedded systems, and capstone delivery.

Research Assistant — NCAI, UET Peshawar, Peshawar, Pakistan | Aug 2019 – Feb 2020

- Developed and deployed CNN/LSTM models for phonocardiogram classification and edge-oriented biomedical signal processing.

EDUCATION

Ph.D., Electrical Engineering — The University of Alabama, Tuscaloosa, AL | May 2024 – Present | CGPA: 3.86/4.0

M.S., Electrical Engineering — FAST National University, Islamabad, Pakistan | 2022 – 2024 | CGPA: 4.0/4.0

B.S., Electrical Engineering — FAST National University, Peshawar, Pakistan | 2014 – 2018 | CGPA: 3.75/4.0

SELECTED PUBLICATIONS

- Khan, F. A., Umar, Z., Jolfaei, A., & Tariq, M. (2024). Explainable AI for epileptic seizure detection in Internet of Medical Things. Digital Communications and Networks.
- Khan, F. A., Umar, Z., Jolfaei, A., & Tariq, M. (2024). Explainable fuzzy deep learning for prediction of epileptic seizures using EEG. IEEE Transactions on Fuzzy Systems.
- K. N. Khan, F. A. Khan, et al. (2021). Deep learning based classification of unsegmented phonocardiogram spectrograms leveraging transfer learning. Physiological Measurement, 42(9).
- Khan, F. A., Jamil, A., Khan, S. A., & Hameed, A. A. (2024). Enhancing robotic manipulator fault detection with advanced machine learning techniques. Engineering Research Express.